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QUESTION: Write a menu-driven C program to perform the following operations using separate functions:

1. Convert a lowercase string to uppercase without using built-in functions.
2. Reverse a string.
3. Check if a string is a palindrome.
4. Exit.

SOLUTION:

1. Aim

To write a menu-driven C program using **separate functions** to perform the following operations on a string:

1. Convert a lowercase string to uppercase without using built-in conversion functions.
2. Reverse a string.
3. Check whether a string is a palindrome.
4. Exit the program.

2. Algorithm

Main Algorithm

1. Start the program.
2. Input a string from the user.
3. Display the menu:
    - Convert lowercase string to uppercase.
    - Reverse the string.
    - Check whether the string is a palindrome.
    - Exit.
4. Accept the user's choice.
5. Use a switch statement to call the corresponding function.

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6. Repeat the menu until the user selects **Exit**.
7. Stop.

Operation 1: Convert Lowercase to Uppercase

1. Start from the first character of the string.
2. Traverse the string until the null character '\0' is encountered.
3. Check whether the current character lies between 'a' and 'z'.
4. If yes, convert it into uppercase using character arithmetic.
5. Continue until all characters have been processed.
6. Display the uppercase string.

Operation 2: Reverse a String

1. Find the length of the string manually.
2. Set one index at the beginning and another at the end of the string.
3. Swap the first and last characters.
4. Move the first index forward and the last index backward.
5. Continue until both indices meet.
6. Display the reversed string.

Operation 3: Check Palindrome

1. Find the length of the string manually.
2. Compare the first character with the last character.
3. Then compare the second character with the second-last character.
4. Continue until the middle of the string is reached.
5. If any pair does not match, the string is not a palindrome.
6. If all corresponding characters match, the string is a palindrome.

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7. Display the result.

C PROGRAM:

```
#include <stdio.h>

// Function declarations
void toUpperCase(char str[]);
void reverseString(char str[]);
void checkPalindrome(char str[]);

int main()
{
   char str[100];
   int choice, i = 0;

   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);

   // Remove newline character added by fgets()
   while(str[i] != '\0')
   {
      if(str[i] == '\n')
      {
         str[i] = '\0';
         break;
      }
      i++;
```

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```
}

do
{
   printf("\n----- STRING OPERATIONS MENU -----\n");
   printf("1. Convert lowercase string to uppercase\n");
   printf("2. Reverse the string\n");
   printf("3. Check whether string is palindrome\n");
   printf("4. Exit\n");

   printf("Enter your choice: ");
   scanf("%d", &choice);

   switch(choice)
   {
      case 1:
         toUpperCase(str);
         break;

      case 2:
         reverseString(str);
         break;

      case 3:
         checkPalindrome(str);
         break;
```

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```
        case 4:
            printf("Program exited successfully.\n");
            break;

        default:
            printf("Invalid choice! Please try again.\n");
    }

  } while(choice != 4);

  return 0;
}



// Function to convert lowercase string to uppercase
void toUpperCase(char str[])
{
  int i = 0;

  while(str[i] != '\0')
  {
    if(str[i] >= 'a' && str[i] <= 'z')
    {
      str[i] = str[i] - ('a' - 'A');
    }
```

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```
        i++;
    }

    printf("\nUppercase string: %s\n", str);
}



// Function to reverse the string
void reverseString(char str[])
{
    int length = 0;
    int i, j;
    char temp;

    // Find length manually
    while(str[length] != '\0')
    {
        length++;
    }

    i = 0;
    j = length - 1;

    // Reverse the string
    while(i < j)
```

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```
  {
    temp = str[i];
    str[i] = str[j];
    str[j] = temp;

    i++;
    j--;
  }

  printf("\nReversed string: %s\n", str);
}



// Function to check palindrome
void checkPalindrome(char str[])
{
  int length = 0;
  int i;
  int palindrome = 1;

  // Find length manually
  while(str[length] != '\0')
  {
    length++;
  }
```

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```
  // Compare characters from both ends
  for(i = 0; i < length / 2; i++)
  {
    if(str[i] != str[length - 1 - i])
    {
      palindrome = 0;
      break;
    }
  }

  if(palindrome == 1)
  {
    printf("\n%s is a palindrome.\n", str);
  }
  else
  {
    printf("\n%s is not a palindrome.\n", str);
  }
}
```

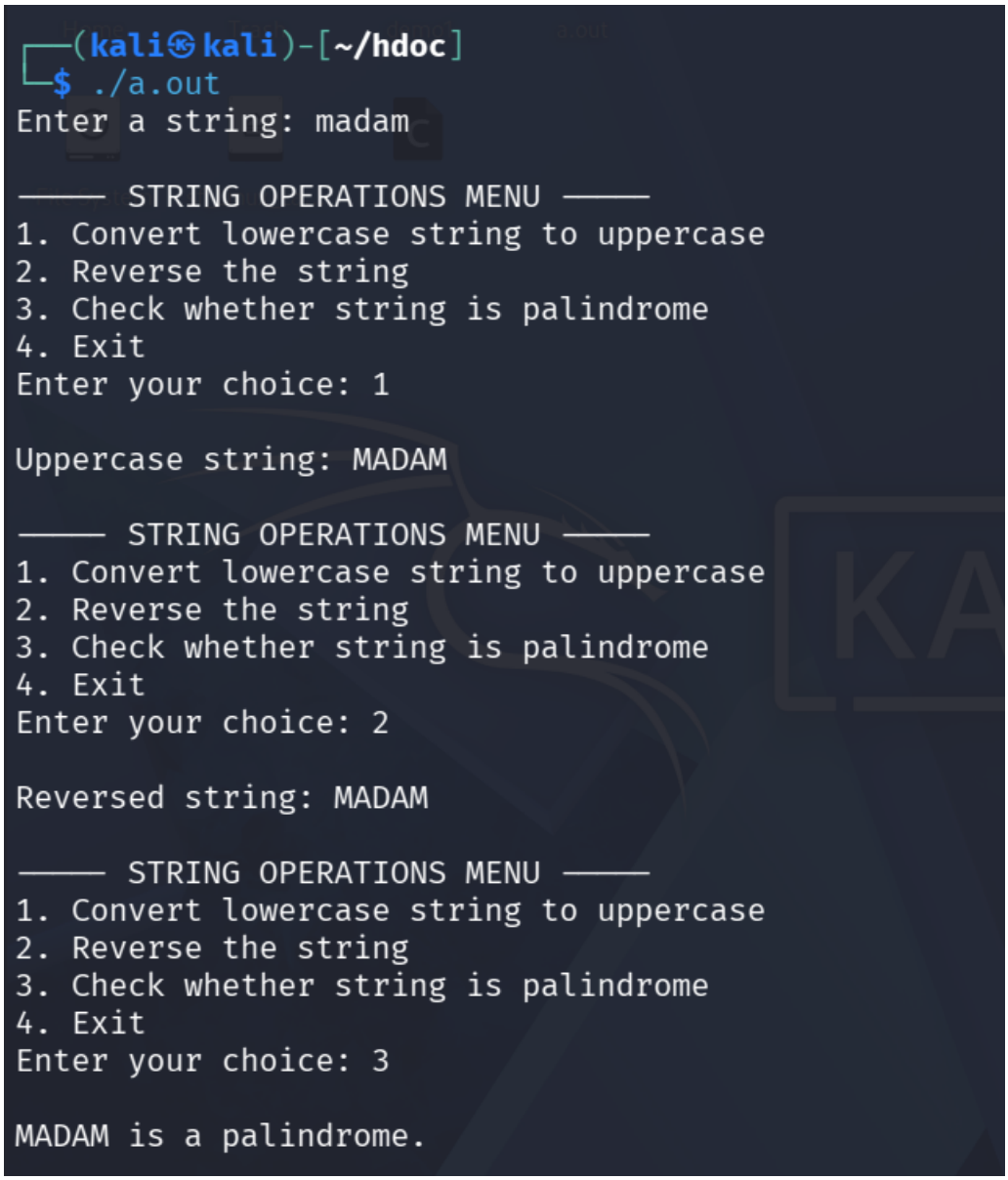
TEST CASE TABLE:

Test Case	Input String	Operation	Expected Result
1	hello	Uppercase	HELLO
2	computer	Reverse	retupmoc
3	madam	Palindrome	Palindrome
4	level	Palindrome	Palindrome

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Test Case	Input String	Operation	Expected Result
5	hello	Palindrome	Not a palindrome
6	abcde	Reverse	edcba
7	programming	Uppercase	PROGRAMMING
8	Any string	Exit	Program terminates
9	Any string	Invalid choice	Displays invalid choice message

COMPILATION AND EXECUTION:



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